

Coherent Errors: Long-Horizon Failure as Action on Misremembered State

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Abstract

Language models degrade over long reasoning chains, and the accuracy drop is usually reported as one number. That number confounds two failures with opposite remedies: the model has lost track of the world, or it holds the world correctly and chooses badly. We separate them in a setting with an exact external state oracle, chess from move notation with no search at inference, by reading linear probes against the true position at every token. Three findings follow. First, aggregate state fidelity, the quantity this literature normally reports, barely predicts behaviour: within a fixed reasoning depth, the correlation between the number of misremembered squares and playing an illegal move ranges from 0.025 to 0.080. Second, the same quantity measured locally predicts it strongly: the probe is wrong on precisely the squares a move touches for 0.806 of illegal moves against 0.384 of legal ones. Third, and centrally, the errors are coherent rather than random. Reconstructing the position the model appears to believe, we find 0.356 of its illegal moves are legal in that believed position, against 0.050 for the same move scored under another position’s belief at equal depth and 0.008 for an arbitrary illegal move. The model is not emitting noise. It is playing correctly on a board it has misremembered. We give the measurement protocol, report a failed prediction and a degenerate metric openly, and evaluate one architectural intervention with it.

Keywords: long-horizon reasoning, state tracking, world models, probing, model diagnostics, design science

1 Introduction

A model that reasons over many steps has to keep track of something. A Transformer stores nothing between positions. Each token’s representation is rebuilt from the whole prefix through attention (Vaswani et al., 2017), so any quantity the task requires, a board position, the value of a variable, which sub-goal remains open, exists only because the network re-derives it, at every layer and every position, from the beginning. That re-derivation is dependable over short spans and unreliable over long ones. Anyone who has watched a model reason for forty steps knows the symptom: it stays fluent and quietly stops being right about its own earlier commitments.

The usual remedies sit outside the model. Tree search grafts deliberation onto the policy (Yao et al., 2023). Process reward models supervise intermediate steps with a learned verifier (Lightman et al., 2024). Self-consistency samples several chains and votes (Wang et al., 2023). Each raises end-task accuracy, and each leaves the original question untouched: what fails inside the network as the chain lengthens? The question matters commercially as well as scientifically, because the two candidate failures recommend opposite purchases. If the model has forgotten the world, a verifier that rescores its outputs is paying repeatedly to repair something the architecture gave away. If the model remembers the world and reasons badly over it, a verifier is exactly right.

Answering the question requires reading an intermediate representation against a known-correct intermediate state, and on open-ended text no such state exists. The intermediate “state” of an argument in prose is not well defined, which is why evaluation in this area is noisy and why claims about it are hard to falsify. Our response is to move the measurement to a domain where the state is exact, cheap, and external, establish what can be learned there, and be explicit about what that does and does not license.

What we find. Working with search-free autoregressive models trained only on chess move notation, we measure the internal state against a rules engine at every token and compare it with what the model actually does. Three results structure the paper.

1. **Aggregate state fidelity is a poor predictor of behaviour.** Probe accuracy over the whole board, which is the standard reported quantity, correlates with illegal-move production at only 0.025 to 0.080 within a fixed depth. A model can have lost a great deal of the board and still play legally, and can have most of it and still fail.
2. **Action-relevant state fidelity is a strong predictor.** Restricting attention to the squares a move actually touches, the probe is wrong for 0.806 of illegal moves and 0.384 of legal ones. What matters is not how much of the world has been lost but whether the part about to be used has been lost.
3. **The errors are coherent.** We reconstruct the board the model appears to believe and ask whether its illegal move is legal there. It is, for 0.356 of illegal moves, with a bootstrap interval of [0.348, 0.364]. The same move scored against another position’s believed board at equal depth is legal 0.050 of the time, and an arbitrary illegal move scored against the model’s own believed board is legal 0.008 of the time. Lossy decoding caps what is detectable at 0.657, the rate for moves that were genuinely legal.

The third result is the one we would defend hardest. Long-horizon failure in these models is not noise breaking through as the context grows. It is competent play in a world that has drifted, which is a claim about memory rather than about reasoning, and it explains why fluency survives correctness: the model is doing the right thing, in the wrong place.

Why chess. Chess presented purely as move notation, with no tree search at inference, gives what prose does not. The true state after any prefix is computable by a rules engine, so probing has a per-token label. Legality is decidable, so a state-tracking failure has an unambiguous observable rather than a judgement call. Move quality is separately scoreable by an engine, so the second failure mode can be measured conditional on the first being absent. Depth is uniformly scaled, unlike “number of reasoning steps” in prose. And the domain is combinatorial, so surface heuristics do not substitute for computation. Chess is our instrument, not our subject, and we are explicit in Section 10 about the reach that buys.

Research questions and objectives.

- **RQ1.** Can long-horizon accuracy loss be separated into state tracking and planning, and estimated separately? **RO1:** define both and give an estimator for each (Sections 4, 6).
- **RQ2.** Does internal state loss explain behavioural failure, or merely accompany it? **RO2:** test the link per token, at the level of the specific state a given action depends on (Sections 7.2, 7.4).
- **RQ3.** How far does a search-free model reason before its state stops being recoverable, and how does that depth respond to scale? **RO3:** measure the horizon across a model ladder against trivial baselines (Sections 7.1, 7.5).

- **RQ4.** Does the diagnostic discriminate between architectural interventions? **RO4:** evaluate a fixed-width recurrent state channel with parameter-matched and ablated controls (Section 7.6).

Contributions.

1. A per-token decomposition of long-horizon error into state-tracking and planning components, with an exact oracle rather than a learned verifier.
2. The finding that aggregate probe accuracy, the standard reported measure of a world model, is close to uninformative about behaviour, while the same measure restricted to action-relevant state is not. This changes what practitioners should measure.
3. The belief-consistency result: long-horizon errors are coherent actions on a decayed state, established against mismatched-belief and random-move controls.
4. An honest account of two methodological failures, a degenerate metric and an unlearnable task design, both of which shaped the protocol and neither of which is usually reported.
5. Transferable design knowledge for building such diagnostics wherever an external state oracle can be constructed.

We present this as design-science research in the sense of Hevner et al. (2004) and Peffers et al. (2007), where the contribution is an artifact together with the design knowledge it carries (Gregor and Hevner, 2013).

2 Related Work

Inference-time search. Deliberation over candidates (Yao et al., 2023), process supervision (Lightman et al., 2024), and sampling with a vote (Wang et al., 2023), built on chain-of-thought prompting (Wei et al., 2022), all buy accuracy with inference compute and take the base policy as given. Our concern is the base policy. A verifier that spends its budget repairing state-tracking errors is compensating for a representational defect at a recurring cost. What we do differently is diagnose which defect is present before recommending a remedy.

Long-context evaluation. Position-dependent retrieval failure (Liu et al., 2024) and effective-context measurement (Hsieh et al., 2024) show usable context is shorter than advertised context. Both measure retrieval of material present in the prefix. Long-horizon reasoning failure is different: the needed quantity was never written down, only implied, and has to be maintained through composition. A model can attend to every token of a sixty-ply game and still not know where the pieces are. What we do differently is measure maintenance rather than retrieval.

World models and probing. Linear probes recovering board state from game transcripts (Li et al., 2023), the linear structure of those representations (Nanda et al., 2023), chess as a state-tracking testbed (Toshniwal et al., 2022), and causal editing of internal associations (Meng et al., 2022) together establish that sequence models form internal world models and that intervening on them changes behaviour. That literature answers whether a world model exists. We take its existence as given and ask a different question: how the fidelity of that world model varies with reasoning depth, and whether its failures explain the model’s behavioural failures at the level of individual actions. Probe design follows the control-task discipline of Hewitt and Liang (2019). To our knowledge the belief-consistency test of Section 7.4, which asks whether a wrong action is right in the model’s own decoded world, has not been reported.

Table 1: Design principles and their realisation.

Design principle	Realisation in this work
DP1. Ground the diagnostic in an external oracle, never in model self-report.	A rules engine labels the exact state after every token, so probe targets are independent of the model being measured.
DP2. Separate the failure modes before attributing any effect.	Planning error is scored only where the move is legal and the state is independently verified as intact.
DP3. Establish what a trivial predictor achieves before claiming a representation exists.	Decodability is reported against a per-square majority predictor, a label-permutation control, and a randomised-weight model of identical shape.
DP4. Measure the state the action actually depends on, not the state in aggregate.	Errors are scored on the squares a candidate move touches, which is what makes the representation predictive of behaviour at all.
DP5. Test coherence, not just correlation.	The believed board is reconstructed and the model’s own action is re-scored in it, against mismatched-belief and random-move controls.

Expressivity of state. Theory bounds what these architectures can track: state-space models are weaker than their recurrent form suggests (Merrill et al., 2024), and intermediate tokens change what a Transformer can express (Merrill and Sabharwal, 2024). That work asks what is representable in principle. We ask what a trained model actually maintains, and where it stops.

Recurrence and state-space models. Structured state-space models (Gu et al., 2022) and selective variants (Gu and Dao, 2024) give constant per-token state, usually motivated by throughput and context length. We use a small supervised recurrent channel not as a contribution in itself but as a test case for whether the diagnostic discriminates between architectures.

Search-free chess models. Policies that reach strong play by distilling engine evaluations without a tree (Ruoss et al., 2024) show the task is learnable without search. Their objective is strength; ours is measurement, so we deliberately study models weak enough to fail. A model that never loses the position teaches nothing about losing it.

3 Design Problem and Requirements

Stated plainly: long-horizon failure is reported as a single number, so it cannot be acted on. A practitioner watching accuracy fall at step thirty cannot tell whether to buy a larger model, add a verifier, or change the architecture, because the measurement does not distinguish the causes. The artifact is a measurement procedure.

In scope. Search-free autoregressive decoders; domains with an exact, cheaply computable state oracle; state of bounded description length.

Out of scope. Inference-time search, which we neither use nor evaluate; tasks whose state grows without bound; and claims about production-scale models, which our compute does not license.

4 Problem Formalisation

Let a task be a token sequence $x_{1:L}$ generated left to right, and $s_t = \Phi(x_{1:t})$ the exact state after the prefix, from an external oracle Φ . For chess this is piece placement, side to move, castling rights, and the en-passant target. The state space is large, but the description length of any s_t is constant in t : a position occupies a few hundred bits however many plies produced it.

Let $h_t^{(\ell)}$ be the residual-stream activation at position t , layer ℓ , and $g^{(\ell)}$ a linear decoder trained to recover s_t from it. Write $\hat{s}_t^{(\ell)} = g^{(\ell)}(h_t^{(\ell)})$ for the decoded, or believed, state.

Definition 1 (State fidelity). $F^{(\ell)}(t) = \Pr[\hat{s}_t^{(\ell)} = s_t]$, the probability the exact state is linearly decodable at depth t .

Definition 2 (Action-relevant fidelity). For an action a touching a set of state components $R(a)$, the probability that $\hat{s}_t$ and s_t agree on $R(a)$. For a chess move this is the origin and destination squares.

Definition 3 (Error decomposition). State-tracking error $E_{\text{state}}(t) = \Pr[\hat{s}_t \neq s_t]$, observable through the illegal-move rate. Planning error $E_{\text{plan}}(t) = \Pr[\text{blunder} \mid \hat{s}_t = s_t]$, from engine centipawn loss conditional on intact state. Observed error is $E_{\text{state}} + (1 - E_{\text{state}})E_{\text{plan}}$, and benchmarks report only that sum.

Definition 4 (Belief consistency). An action is belief-consistent when it is invalid in s_t but valid in $\hat{s}_t$. The rate of belief consistency among invalid actions distinguishes coherent action on a wrong state from incoherent action.

5 The State-Space Bottleneck

Proposition 1 (Cost mismatch). For an attention-only decoder, computing $h_t^{(\ell)}$ attends over t prior positions, so maintaining state costs $O(t)$ reads and $O(t)$ key-value memory while the state has description length $O(1)$. Nothing amortises it: s_t and s_{t+1} differ by one move and are computed independently.

Two consequences were testable and we tested both. Fidelity should fall with depth even well inside the context window, because the re-derivation circuit must compose more updates within a fixed layer budget. It does (Section 7.1).

We also predicted fidelity would peak at intermediate layers and fall toward the output, the state being a means rather than an end. **That prediction is wrong.** Decodability rises through the stack and is maximal at the final block, from 0.322 at the first layer to 0.543 at the last. For this task the state sits close to the target: the next legal move is a direct function of the position, so a representation making the position linearly available is also a good representation for the output head. We keep the failed prediction because it constrains where a state carrier should be inserted, and our spacing was chosen before we knew.

6 Diagnostic Protocol

Data. Public game records above a rating floor, deduplicated by move-sequence hash, split by *game* so no prefix is shared between probe fitting and evaluation. The corpus holds 120,000 games: 96,000 for training, 12,000 for probes, 12,000 for evaluation, mean length 78.2 plies, move vocabulary 1,926.

Table 2: Chess arm. Occupied-square state accuracy (occ), illegal top-1 move rate, and behavioural horizon H_{ill} (first ply bucket where the illegal rate exceeds the threshold). Mean $\pm$ s.d. over seeds.

Model	Params	seeds	occ@40–50	illegal@40–50	$H_{\text{ill}}(.05)$	$H_{\text{ill}}(.10)$
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Oracle. A rules engine replays every game and emits the exact state after every token, verified complete before any model is measured.

Probes. One linear head per layer, fitted on the probe split and reported on the evaluation split. Because most squares are empty, raw per-square accuracy is misleadingly high, so we report accuracy on occupied squares against a per-square, per-depth majority predictor.

Controls. A majority predictor; a label-permutation control pairing activations with another game’s state at equal depth, preserving marginals while destroying the relation to the input; and a randomised-weight model of identical architecture, bounding what probe capacity alone extracts.

Behavioural measures. Illegal top-1 move rate against depth, and the probability mass on legal continuations, both read against the oracle.

7 Results

7.1 State and behaviour both decay with depth

Occupied-square decodability falls from 0.974 in the opening to 0.063 in the deepest bucket, while the illegal top-1 rate rises from 0.0004 to 0.319 and the probability mass on legal moves falls from 0.993 to 0.455. At mid-game the trained model is well clear of every trivial explanation: 0.559 against 0.261 for the majority predictor, ?? for the label-permutation control, and ?? for a randomised-weight model of identical shape. There is a real state representation, and it is really being lost.

7.2 Aggregate fidelity does not explain behaviour

The two curves fall together, and it would be easy to stop there and call them one phenomenon. They are not, at least not in the form usually reported. Within a fixed depth bucket, so that both quantities cannot simply be tracking depth, the correlation between the number of misremembered squares and whether the model plays an illegal move ranges from 0.025 to 0.080. Aggregate board fidelity, the number this literature reports as evidence of a world model, carries almost no information about whether the next action will be valid.

We think this deserves emphasis because it is a negative result about a standard practice. Reporting that a probe recovers a world model at some accuracy says little about whether that model is load-bearing.

7.3 Action-relevant fidelity does

Restricting the measurement to the squares a move touches changes the picture. The probe is wrong on those squares for 0.806 of illegal moves and 0.384 of legal moves. The relevant question is not how much of the world the model has lost but whether it has lost the part it is about to use.

7.4 The errors are coherent

If state loss drives the failure, an invalid action should be valid in the state the model believes it is in. We reconstruct that believed position from the probe and re-score the model’s own move in it, clearing castling rights so nothing is licensed by an assumption we did not decode, and testing pseudo-legality because a decoded board may lack a king.

Of the illegal moves the model plays, 0.356 are legal in its own believed position, with a bootstrap interval of $[0.348, 0.364]$. Two controls bound the alternatives. The same move scored against a believed position decoded elsewhere at equal depth is legal 0.050 of the time, so the effect is specific to this belief at this position rather than to plausible boards generally. An arbitrary illegal move scored against the model’s own believed position is legal 0.008 of the time, so the believed board is not permissive. The ceiling on what this test can detect is 0.657, the rate at which genuinely legal moves are recovered as legal in the believed board, which is below one because probe decoding is lossy.

Against that ceiling, roughly half of the model’s illegal moves are coherent actions in a misremembered world, and the enrichment over the matched control is a factor of 7.0. This is what we mean by coherent error, and it is the clearest statement we can make that long-horizon degradation here is a memory failure rather than a reasoning failure.

7.5 Scale

7.6 An intervention, evaluated by the diagnostic

The diagnostic is only useful if it discriminates. We test a fixed-width gated recurrent channel inserted after every k -th attention block, carrying $z_t \in \mathbb{R}^{d_s}$ with $d_s \ll d$:

$$u_t = W_{\text{in}} h_t, \quad \alpha_t = \sigma(W_g[h_t; z_{t-1}]), \quad z_t = (1 - \alpha_t) \odot A z_{t-1} + \alpha_t \odot \phi(u_t),$$

read back as $h_t \leftarrow h_t + W_{\text{out}} z_t$, with A diagonal so inference-time state is $O(1)$ per token, and an optional auxiliary loss $\lambda \mathcal{L}_{\text{state}}$ supervising z_t to be the state. We separate architecture ($\lambda = 0$), supervision ($\lambda > 0$), and read-out (zeroing W_{out} at test time), against a parameter-matched attention-only control whose feed-forward block absorbs the same parameter count.

7.7 A degenerate metric, declared

Our horizon was pre-registered on exact-position fidelity. That quantity reaches zero by roughly ply 0 in every condition, so it cannot discriminate between them, and conditioning on it is impossible because exact states essentially do not occur past the opening. We therefore report horizons on occupied-square accuracy and on the illegal-move rate. The revision was made after inspecting baseline data and before evaluating any intervention, and every definition is applied unchanged across conditions. We flag it rather than presenting the revised metric as the original plan, and we note that anyone measuring state tracking with an all-or-nothing criterion will obtain an unusable number for the same reason.

8 Discussion

The practical content of this paper is a change in what to measure. Reporting that a probe recovers a world model at some aggregate accuracy is weak evidence that the model is load-bearing, because aggregate accuracy is nearly uninformative about the next action. Measuring the same

representation on the components an action depends on is informative. Any team instrumenting a system for long-horizon reliability should condition on the state the action uses.

The belief-consistency result reframes the failure. A model that plays a coherent move in a misremembered position is not behaving erratically, and treatments that assume erratic behaviour are aimed at the wrong target. Sampling several chains and voting will not help if every chain is drawn from the same drifted state. A verifier that scores outputs is well placed to catch the error and badly placed to prevent it, and it will pay that cost on every query for as long as the system runs.

The protocol generalises past chess. Anywhere an external checker can reconstruct the intended intermediate state, program execution traces, spreadsheet recalculation, structured database updates, ledger reconciliation, the same measurements are available. The requirement is not a game. It is an oracle.

We also treat the negative results as part of the contribution. A predicted layer profile came out backwards. A pre-registered metric turned out to be degenerate. A first design of a second domain turned out to be unlearnable, for a reason that was diagnosable from the loss curve. Each changed a decision, and each is reported where it did so.

9 Implications for Individuals, Organisations, and Society

For the people who rely on long answers. Someone working through a discharge summary, a tenancy agreement, or a benefits calculation with a model’s help is rarely able to audit step twenty-eight. Our central finding says something uncomfortable about that situation: when state has drifted, the output is not merely wrong, it is coherent, and coherence is exactly the signal a non-expert reader uses to decide whether to trust it. The failure is therefore least visible to the reader least able to check it. A measured horizon, stated plainly, gives that reader something they do not currently have.

For managers and organisations. Long-horizon reliability is a property to be governed rather than a number to be procured. The decomposition tells a team which remedy fits: a recurring inference-time cost for a policy problem, or a one-time structural change for a memory problem. It also supports ordinary practice, triage of workloads that exceed a measured horizon, human review placed at the depth where reliability falls rather than spread evenly, and audit evidence that a limit was known and respected.

Strategic implications. An organisation that can measure its own horizon is better placed than one accepting a vendor’s average. The diagnostic needs a rules engine and a linear probe, not privileged access at scale, so it can be owned and audited in-house. Treating a measured limit as documentation rather than as an embarrassment turns a weakness into a defensible position with auditors and regulators.

For society and digital equity. Buying reliability with inference-time compute concentrates capability among those who can afford to spend it on every query. Diagnosing the failure instead, and fixing what is structural, is the kind of remedy that travels to on-device assistants and public-sector systems that cannot buy deliberation per request. The equity question is not whether frontier systems can reason for fifty steps. It is whether affordable ones can be honest about where they stop.

10 Limitations

- L1 Scale.** Our models are small and trained on a modest corpus, far below the chess-specific literature and any frontier system. Undertrained models are easier to study because they fail within a measurable range, which is what makes the instrument work, but the horizon values describe these models. Section 7.5 estimates the trend rather than assuming it away.
- L2 One domain family.** Chess carries an exact oracle by construction, which is what makes it measurable and also what makes it unrepresentative of open-ended reasoning.
- L3 Probe decoding is lossy.** The believed board is an approximation, which is why we report the legal-move recovery rate as an explicit ceiling rather than treating the decoded state as ground truth.
- L4 Pseudo-legality.** Belief consistency is tested with pseudo-legal moves, since a decoded board may lack a king and make check tests ill-defined. This is a permissive criterion, which the random-move control is designed to bound.
- L5 Probes show availability, not use.** Linear decodability shows information is present and linearly available. The belief-consistency result is much stronger evidence of use than correlation would be, but it remains observational rather than interventional.
- L6 A metric was revised after seeing data.** Declared in Section 7, applied uniformly, reported alongside the original.
- L7 No natural-language transfer test.** Establishing that these findings hold for mathematical or textual reasoning needs matched general-corpus pre-training, which our compute did not cover. It is absent rather than approximated.

11 Avenues of Future Research

- FR1** Extend the scale ladder far enough to estimate how horizon grows with parameters and data, and test whether the belief-consistency rate is scale-invariant even where the horizon moves.
- FR2** Replicate the decomposition in domains that were not designed as benchmarks, such as program execution or ledger reconciliation, where an oracle exists but the data is not constructed.
- FR3** Run the natural-language transfer test with matched pre-training, the experiment that decides whether this line of work reaches practice.
- FR4** Move from observation to intervention, editing action-relevant state directions and measuring whether the induced action follows the induced belief.
- FR5** Develop state supervision that does not require an oracle, for example from execution traces or agreement between sampled continuations.
- FR6** Use action-relevant fidelity as a live confidence signal, abstaining or escalating when the state an action depends on is poorly determined.
- FR7** Pre-register horizon metrics on pilot data, so metric selection is settled before confirmatory runs rather than declared afterwards.

12 Conclusion

Long-horizon failure in Transformers is normally measured as one number and treated with more inference-time compute. Measured against an exact oracle it separates into two, and in the setting we can measure cleanly, the memory component dominates and behaves in a specific way. Aggregate world-model accuracy says little about what a model will do next. The same measurement, restricted to the state an action depends on, says a great deal. And the failures themselves are

coherent: the model plays a reasonable move in a board it has misremembered. The people who stand to gain from measuring this are not primarily the builders of frontier systems. They are the readers who cannot check step twenty-eight, and the organisations that must decide, with evidence, how far to let a model reason before a person looks.

Declarations

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